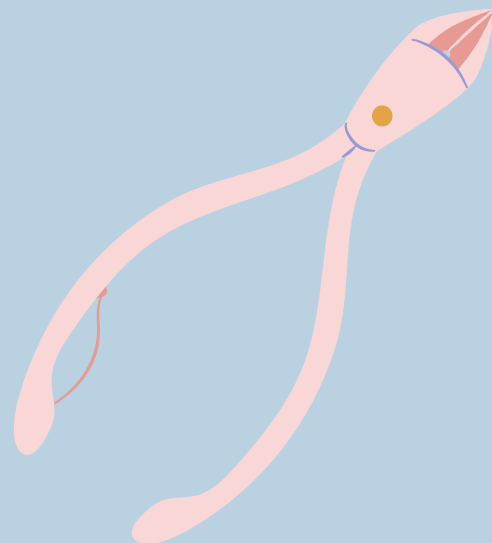
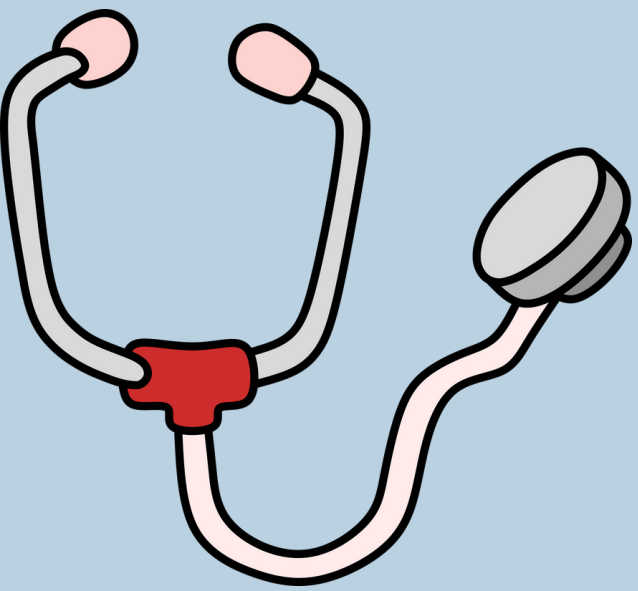


OVARIAN CYST IN ADOLESCENCE AND ITS COMPLICATIONS

***Presented by Esra yadgar,
Hadiya Hamasharif, Shanyar Nawzad***



PROJECT OVERVIEW

- *Introduction*
- *Pathophysiology*
- *Risk factor*
- *Clinical features*
- *Investigation*
- *Management*
- *complication*
- *Follow up*
- *conclusion*
- *references*



INTRODUCTION

- *Fluid-filled sacs that develop within or on the surface of the ovary*
- *May be functional (physiological) or pathological*
- *Very common in early reproductive years*
- *Most are benign and linked to normal menstrual physiology*
- *Adolescents have immature HPO axis - irregular ovulation .*
- *High chance of functional cyst formation*
- *Common soon after menarche*
- *Pathological cysts less common but clinically significant*

EPIDEMIOLOGY OF OVARIAN CYSTS IN ADOLESCENTS

- *Incidence of 1-2%.*
- *Peak prevalence at age 15: 31.3%*
- *Children 5-9 years: 1.8%*
- *Adolescents 10-18 years: 18.8%*

TYPES OF OVARIAN CYSTS IN ADOLESCENTS

A. Functional Cysts (Most common)

1. Follicular cyst

- Failure of a dominant follicle to rupture.*
- Usually <6 cm, asymptomatic.*

2. Corpus luteum cyst

- After ovulation, corpus luteum seals and fills with fluid/blood.*
- May cause delayed menses and unilateral pain.*

3. Theca lutein cysts

- Large, bilateral; associated with (pregnancy, molar pregnancy).*

CONT.

B. Pathological Cysts

1. Benign germ cell tumors

- *Dermoid cyst (Mature cystic teratoma) - most common pathological cyst in adolescents.*

2. Benign epithelial tumors

- *Serous cystadenoma*
- *Mucinous cystadenoma*

3. Endometrioma ("Chocolate cyst")

- *Due to endometriosis; pelvic pain, dysmenorrhea.*

4. Paraovarian cysts

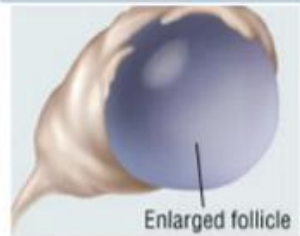
- *Not from the ovary itself; arise from remnants of Wolffian ducts.*

Types of Ovarian Cysts in Adolescents

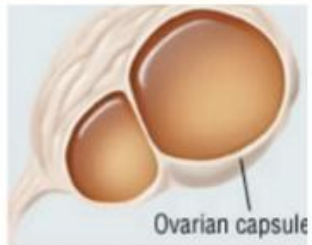
Functional Cysts

(Most common)

Follicular cyst



Corpus luteum cyst

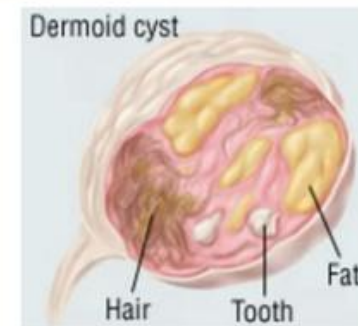


Theca lutein cysts

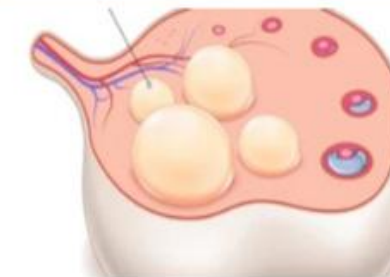


Pathological Cysts

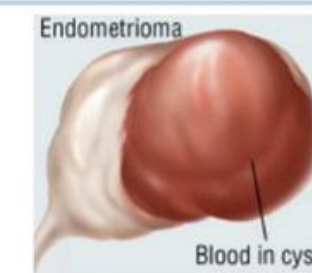
Dermoid Cyst (Mature Cystic Teratoma).



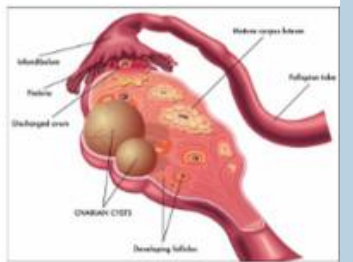
Benign Epithelial Tumors



Endometrioma (Chocolate Cyst)



Paraovarian Cysts



GENERAL PATHOPHYSIOLOGY OF OVARIAN CYSTS

- *Ovulatory cycle disruption prevents normal follicle maturation or regression.*
- *Immature/imbalanced HPO axis → follicles fail to mature, rupture, or involute.*
- *Fluid or blood accumulates inside ovarian structures → cyst formation.*
- *Excess FSH, LH, or hCG overstimulates ovarian cells → enlarged cystic structures.*
- *Ovaries are highly vascular, so bleeding can also create cystic spaces.*
- *Non-hormonal growths (epithelial, germ cell, or ectopic endometrial tissue) can form persistent cysts.*
- *Final result: a fluid-filled sac forms from abnormal growth, failed ovulation, or failed regression.*

CLASSIFICATION OF OVARIAN CYSTS

Type of Cyst	Mechanism / Pathophysiology	Key Features
Follicular Cyst	Failure of LH surge → follicle doesn't rupture → continues to grow under estrogen	Most common; thin-walled; may cause hyperestrogenism
Corpus Luteum Cyst	Corpus luteum doesn't regress or bleeds into itself; highly vascular	Delayed menses; may be hemorrhagic; common in pregnancy
Theca Lutein Cyst	Excess β -hCG overstimulation of theca interna	Bilateral, large; seen in GTD & multiple gestation
Dermoid Cyst (Teratoma)	Arises from totipotent germ cells differentiating into multiple tissues	Hair, teeth, fat; high torsion risk
Serous/Mucinous Cystadenoma	Surface epithelium proliferation	Mucinous: risk of pseudomyxoma peritonei
Endometrioma	Recurrent bleeding of ectopic endometrial tissue	"Chocolate" fluid; dysmenorrhea; infertility
Paraovarian Cyst	Arises from Wolffian duct remnants	Adjacent to ovary; not hormonally active

RISK FACTORS FOR OVARIAN CYSTS IN ADOLESCENCE

Risk Factors for Ovarian Cysts in Adolescence

1 Hormonal Factors



Adolescence is marked by significant hormonal fluctuations as the HPO axis matures

Increased gonadotropin levels stimulate ovarian follicles

2 Early or Irregular Menstrual Cycles



Early menarche

Irregular menstrual cycles

Prolonged cycles (oligomenorrhea)

4



Use of Fertility or Ovulation-Induction Medications

- Drugs such as clomiphene citrate can stimulate ovaries, increasing the development of multiple or ovarian cysts.

3 Polycystic Ovary Syndrome (PCOS)



One of the most common causes of ovarian cysts in adolescents

Characterized by chronic anovulation, hyperandrogenism, and multiple small ovarian follicles

5 Endometriosis



Growth of endometrial tissue outside the uterus may lead to endometriomas (chocolate cysts) on the ovaries

6 Hypothyroidism



Thyroid hormone deficiency interrupts the HPO axis

Leads to menstrual irregularities and chronic anovulation

Increases the likelihood of persistent functional ovarian cysts

7 Genetic and Family History



Family history of ovarian cysts, PCOS, endometriosis, or ovarian cancer

BRCA1 and BRCA2 genetic mutations primarily raise risk of ovarian tumors

8 Pelvic Inflammatory Disease (PID) and Pelvic Infections



PID can lead to tubo-ovarian abscess formation

Often associated with sexually transmitted infections

CLINICAL FEATURES, SIGNS, AND SYMPTOMS

- *Many ovarian cysts in adolescents are **asymptomatic**, particularly when they are small, functional, and short-lived.*
- *Because they do not compress surrounding structures or impair ovarian function, they commonly go unnoticed.*
- *Most asymptomatic cysts are discovered incidentally during imaging performed for unrelated medical evaluations.*

CONT.

When symptoms occur, their nature depends on the cyst's size, type, and complications.



1 - Pelvic or Lower Abdominal Pain

- The most common presenting symptom.*
- Reported in $\approx 60\%$ of symptomatic adolescents.*
- Character of pain: (Dull or sharp , Intermittent or constant , Unilateral or bilateral*

2 - Sudden, sharp mid-cycle pain *felt in the lower abdomen due to follicular rupture during ovulation*

CONT.

3 - Menstrual Irregularities

- *Irregular menstruation*
- *Oligomenorrhea, menorrhagia*
- *Variations in cycle length*

→ *Often related to hormonal fluctuations associated with functional cysts.*

4 - As cysts enlarge, they may exert pressure on adjacent organs, resulting in:

- *Bloating or abdominal fullness*
- *Pelvic pressure sensation*
- *Increased urinary frequency due to bladder compression*
- *Occasional nausea or vomiting*
- *Chronic pelvic discomfort due to mass effect*



INVESTIGATION OF OVARIAN CYSTS IN ADOLESCENTS

- ***Pelvic Ultrasonography — First-Line Test***

Ultrasound is the primary investigation because it is non-invasive, radiation-free, and evaluates the cyst's size, location, morphology, and internal contents.

 ***A transabdominal Ultrasound*** (with a full bladder) is preferred for non-sexually active adolescents.

 ***Transvaginal Ultrasound*** may be used in sexually active patients for better detail.

 ***Doppler Ultrasound*** : increased flow may suggest a neoplasm, absent or low flow may suggest torsion.

CONT.

- ***MRI*** : This is used if the ultrasound results are unclear. It provides a more soft-tissue detail and without radiation, which is helpful for complex cysts.
- ***CT Scan***: This is rarely used. It is reserved for emergency situations to quickly assess for other abdominal problems.
- ***Laboratory Tests***
 - Pregnancy test and β -hCG ,CBC
 - Tumor markers (CA-125, AFP, LDH)
 - Hormonal assays
- ***Diagnostic Laparoscopy***

OVARIAN CYSTS:
BENIGN VS
MALIGNANT
CHARACTERISTICS

Parameter	🟢 Benign Cysts	🔴 Malignant Cysts
📏 Size & Growth	< 8 cm Stable size	> 10 cm Rapid enlargement
🏰 Wall Structure	Thin walls (< 3 mm) Smooth contours	Thick walls (> 3 mm) Irregular shape
🔬 Internal Architecture	Anechoic No septations No solid areas	Thick septations Solid components Papillary projections
💗 Vascular Pattern	Avascular Peripheral flow only	Hypervascular Central blood flow Low resistance
🧪 Biomarker Profile	Normal CA-125 Stable markers	Markedly elevated CA-125 Rising trend

MANAGEMENT OVERVIEW & NON-SURGICAL OPTIONS

- *Expectant Management (Watchful Waiting)*
 - *Preferred for: Uncomplicated, asymptomatic cysts.*
 - *Method: Regular monitoring via ultrasound.*
 - *Small functional cysts often regress spontaneously within 8 weeks.*
- *Medical Management*
 - *Hormonal Therapy (e.g., Oral Contraceptives) containing 30 µg of ethinyl estradiol can be considered to suppress the hypothalamic-pituitary-ovarian axis to prevent development of physiologic cysts in the future*
 - *can be considered to reduce recurrence of functional cysts but does not resolve existing cysts.*
 - *Symptomatic Relief*
 - *Analgesics for pain management.*

WHEN IS SURGERY NEEDED?

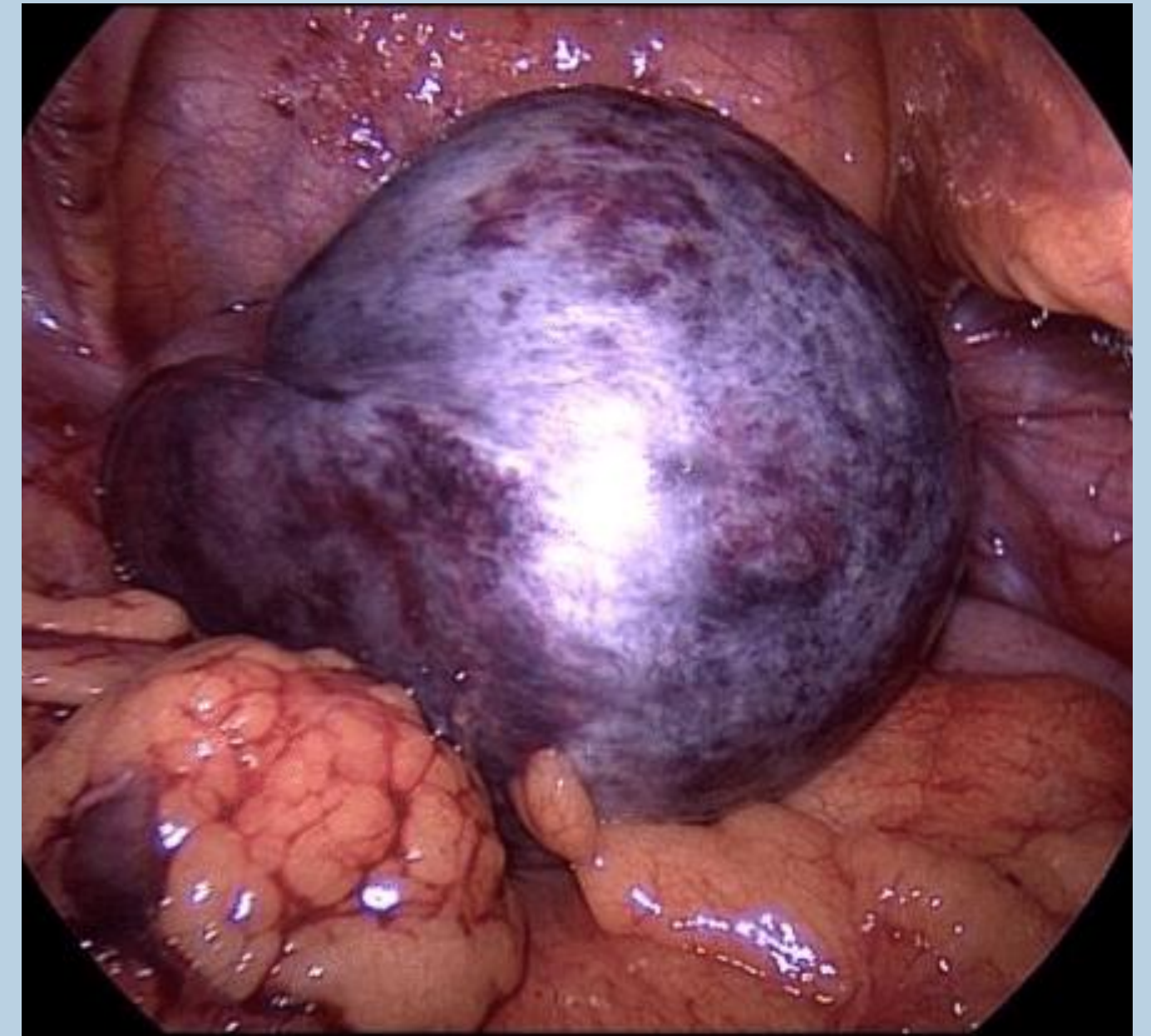
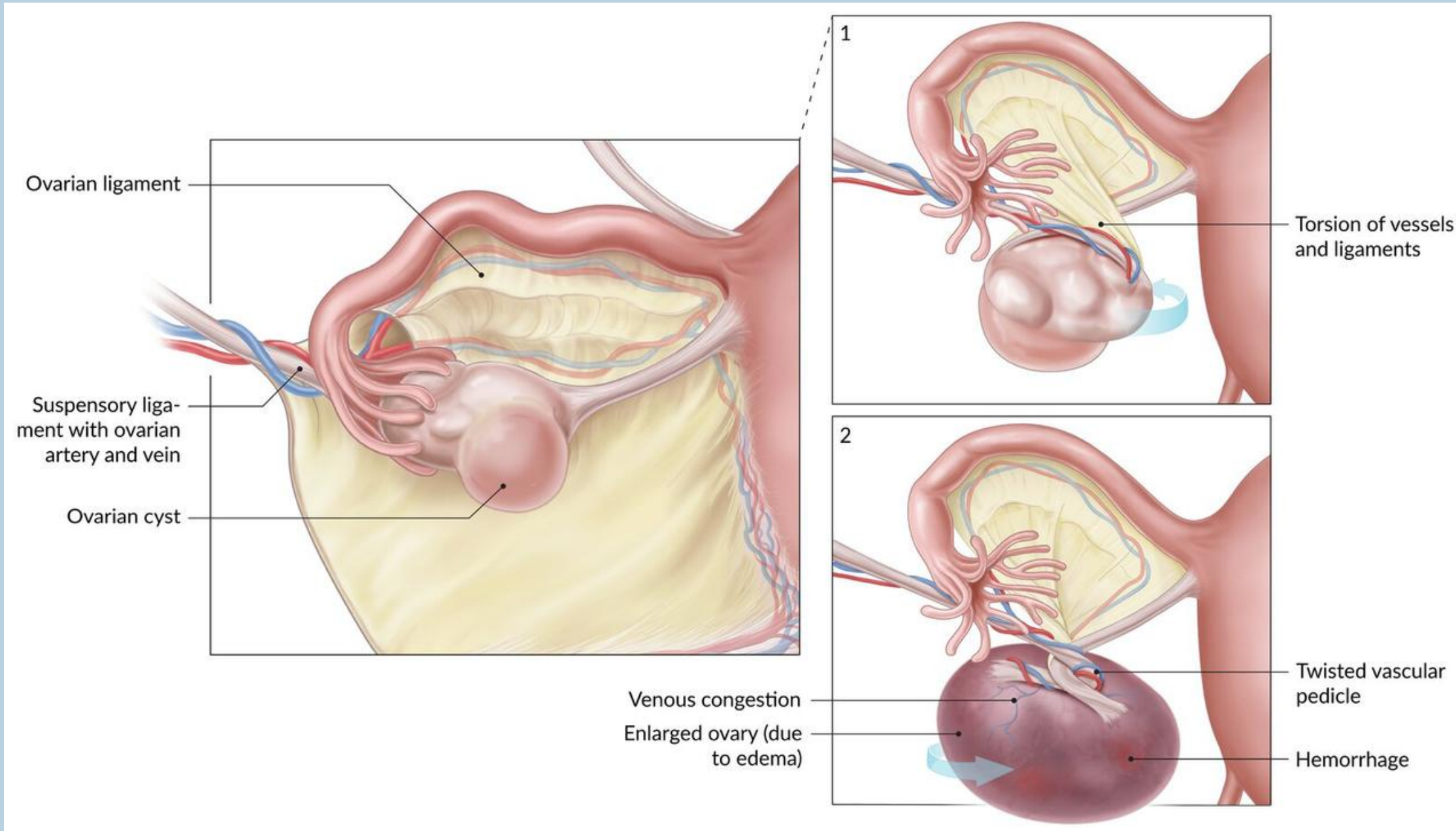
- *Indications for Surgical Intervention:*
 - *large cysts (>6)*
 - *Cysts that are persistent or growing.*
 - *Complicated Cysts: Suspicion of torsion, rupture, or malignancy.*
 - *Cysts causing significant symptoms.*



CONT

- *Preferred Surgical Approach: Laparoscopy*
 - *Advantages: Less pain, shorter hospital stay, fewer adhesions.*
- *Primary Procedure: Cystectomy is usually preferred*
 - *Removal of the cyst while preserving the ovary.*
 - *Goal: Preserve fertility and hormonal function.*
 - *Oophorectomy (removing the entire ovary) is a last resort.*

OVARIAN TORSION



COMPLiCATIONS: OVARIAN TORSiON

- *Twisting of the ovary, cutting off blood supply. A surgical emergency.*
 - *Sudden, severe unilateral pelvic pain.*
 - *Nausea, vomiting, sometimes fainting.*
 - *The gold standard in the laparoscopic approach (distortion , cystectomy with ovarian preservation).*
 - *Critical Principle: Preserve the ovary always, even if it looks damaged. It often recovers function.*
 - *Do not remove the ovary (oophorectomy) based on appearance.*

COMPLICATIONS: RUPTURE & INFECTION

- Cyst Rupture & Hemorrhage
 - Sudden, severe pelvic pain; dizziness; drop in hemoglobin.
 - Management:
 - Stable: Conservative (pain meds, monitoring).
 - Unstable: Laparoscopy to aspirate blood and preserve the ovary.
- Infection & Tubo-Ovarian Abscess (TOA)
 - A serious complication of Pelvic Inflammatory Disease (PID).
 - Can lead to chronic pelvic pain and infertility if untreated.
 - Requires prompt antibiotic treatment and sometimes drainage.

FOLLOW-UP & KEY TAKEAWAYS

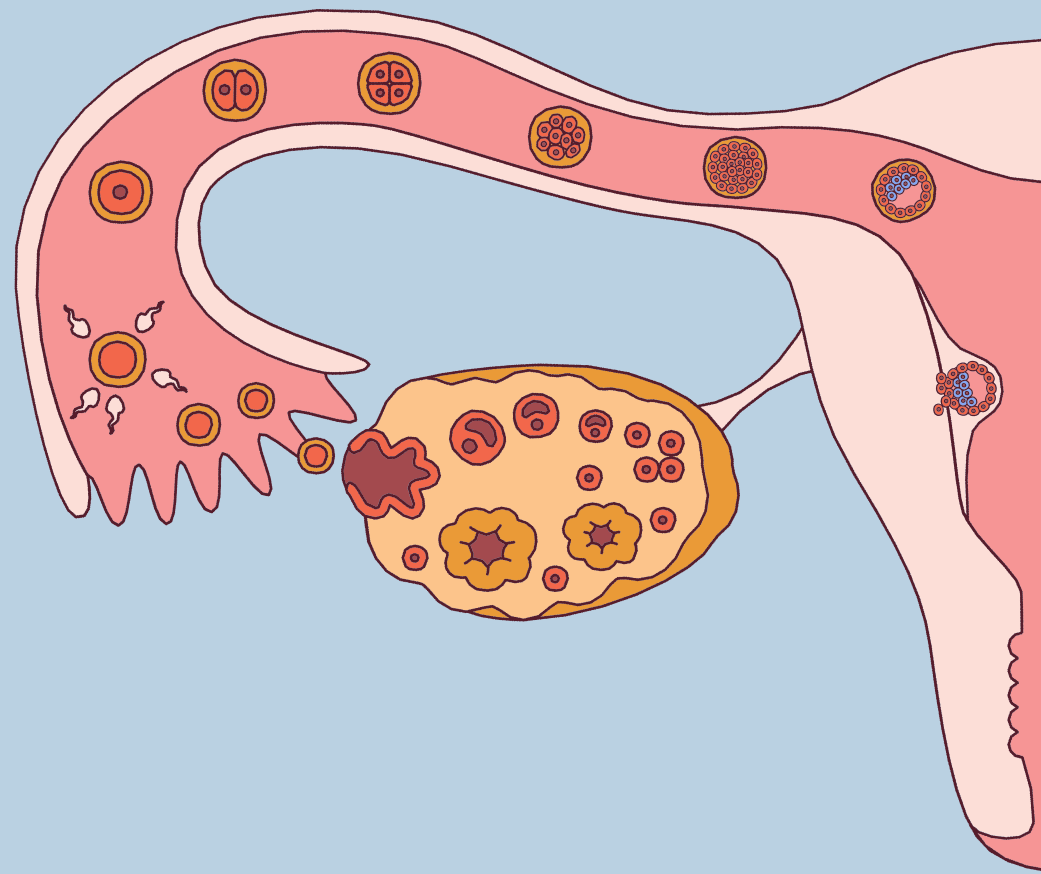
- Regular Monitoring:
 - Ultrasound follow-up until resolution or stability.
 - Close clinical monitoring for recurrence, especially in the first year.
- Patient Education is Crucial:
 - Educate on warning signs: severe pain, fainting, fever.
 - These symptoms require urgent medical care.
- Multidisciplinary Care:
 - Often needed for complex or recurrent cases to ensure the best outcome for the adolescent's health and future fertility.

CONCLUSIONS

- Adnexal lesions in adolescents commonly represent functional ovarian cysts, which can often be managed expectantly.
- Ovarian neoplasms are also common and are most often benign.
- Careful history, physical exam, and ultrasound imaging is key to achieve diagnosis.
- When surgical intervention is indicated, ovarian-sparing surgery is preferred when pre-operative investigations suggest a benign etiology.
- Further efforts are needed to improve diagnostic algorithms to prevent unnecessary oophorectomies within this patient population.

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